ORGANIZATION OF LIVING SYSTEM.

THE MACROMOLECULES

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LEARNING OBJECTIVES

At the end of the lecture the student should be able to:

- Understand the philosophy and meaning of life
- Define level of organization
- List the various levels of organizations
- Describe Chemical organization of the human body
- Describe macromolecules
- Relate macro-molecules with different functions

All religions, arts and sciences are branches of the same tree

Albert Einstein

PHILOSOPHY OF LIFE

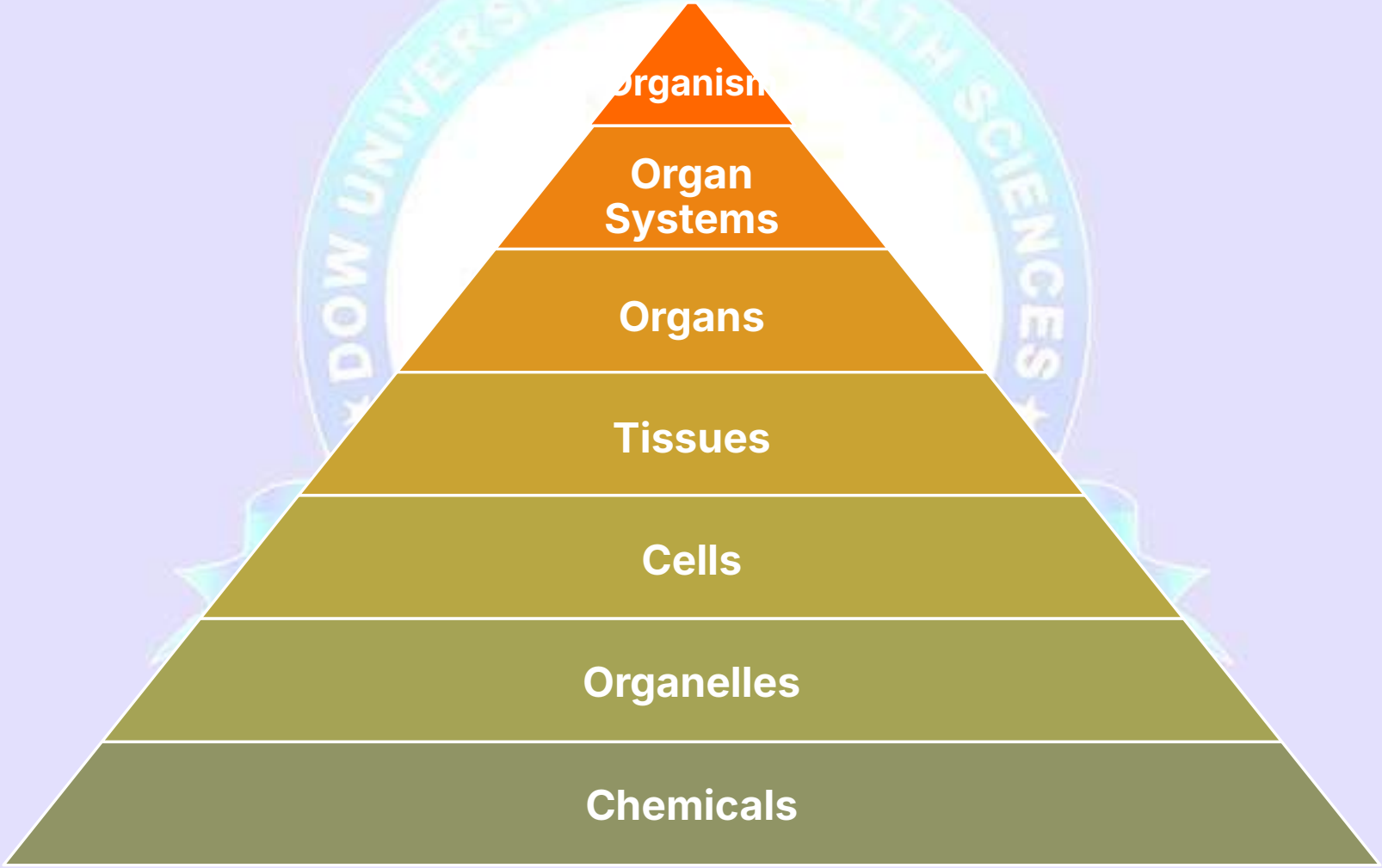
- Period between birth and death
- Which will be questioned
- Learn, understand and Implement
- Levels of organization for healthy living in both the worlds

LEVELS OF ORGANIZATION

The steps in evolution

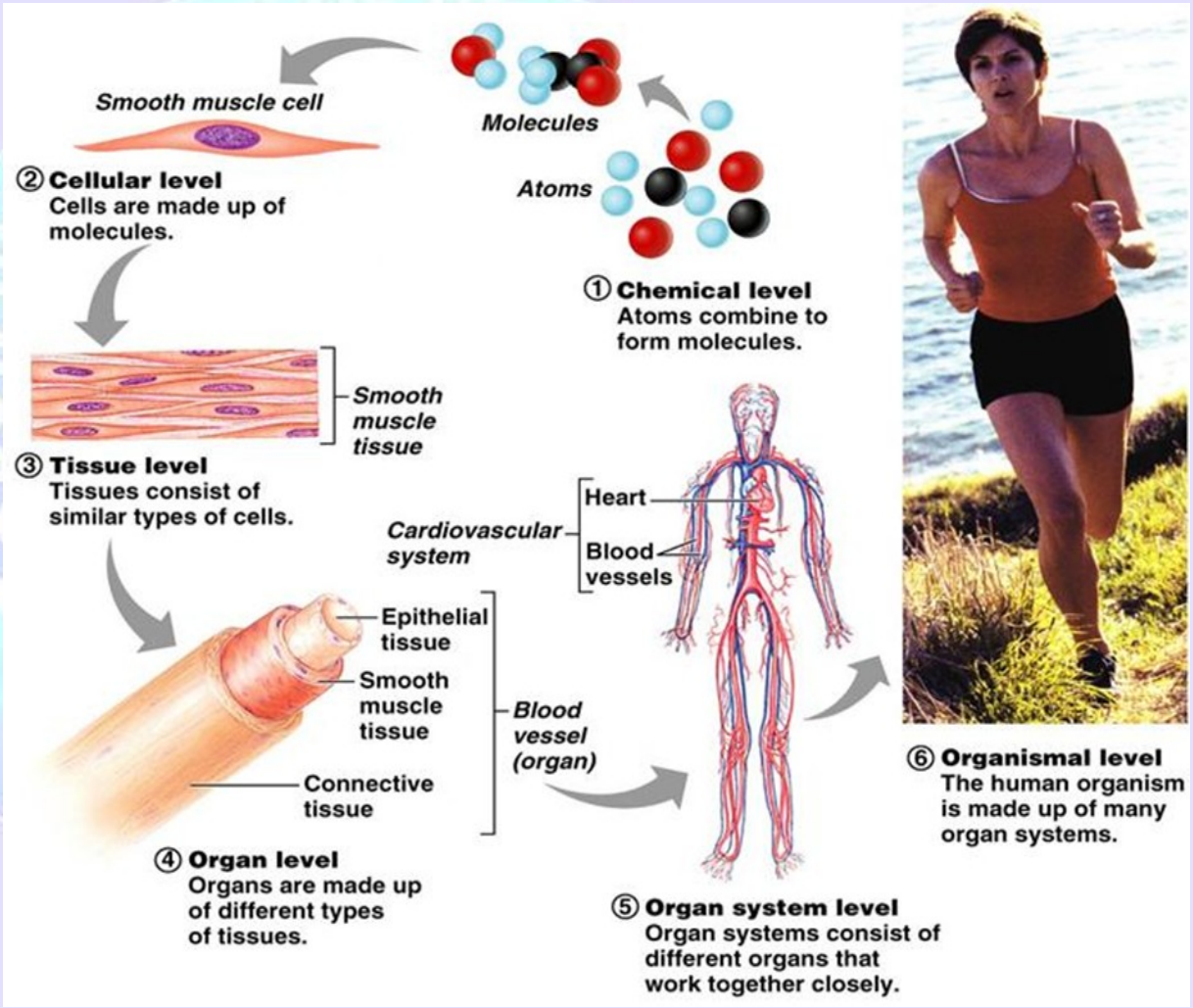
- Structural
- Functional
- Biochemical

STRUCTURAL ORGANIZATION OF LIVING BODY

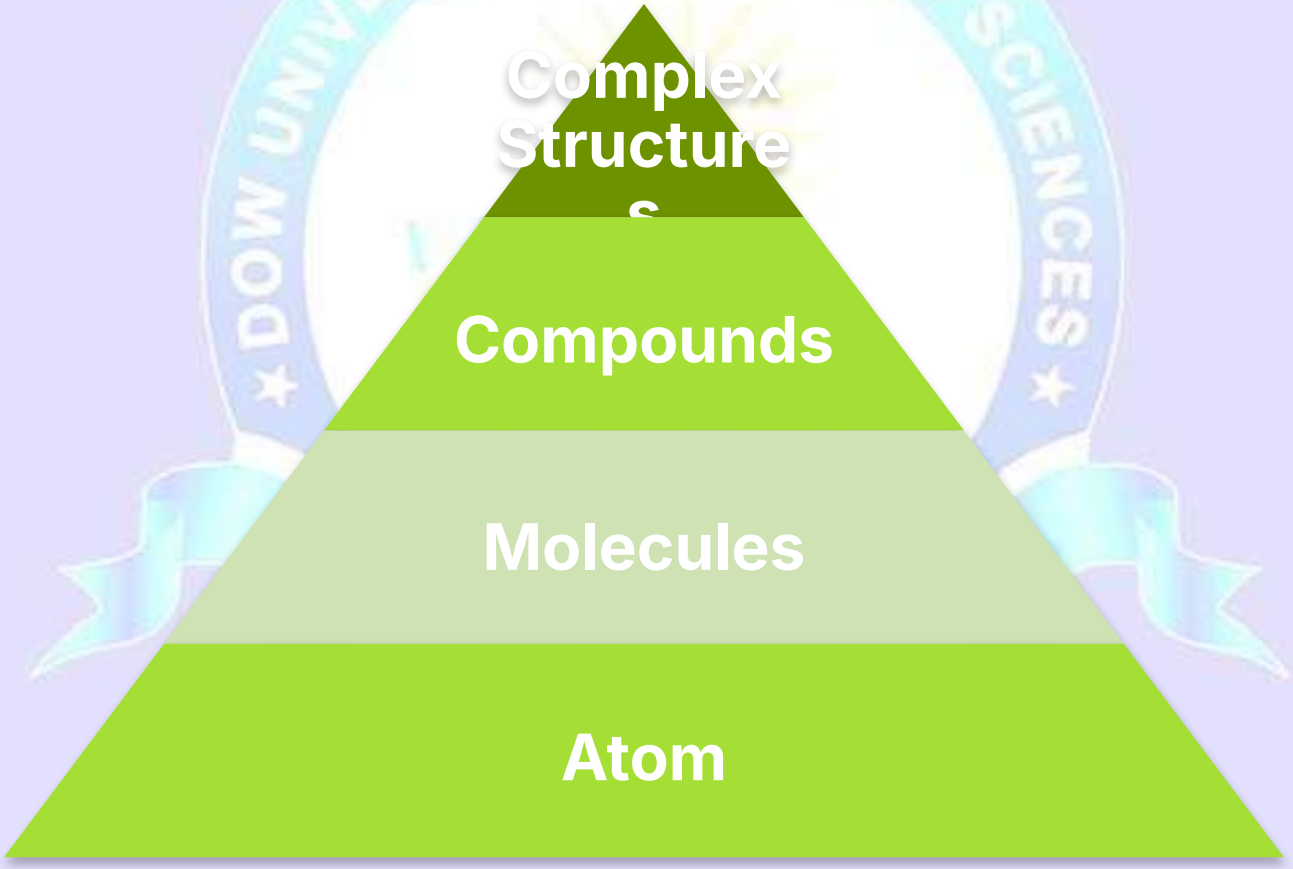


FUNCTIONAL LEVELS OF

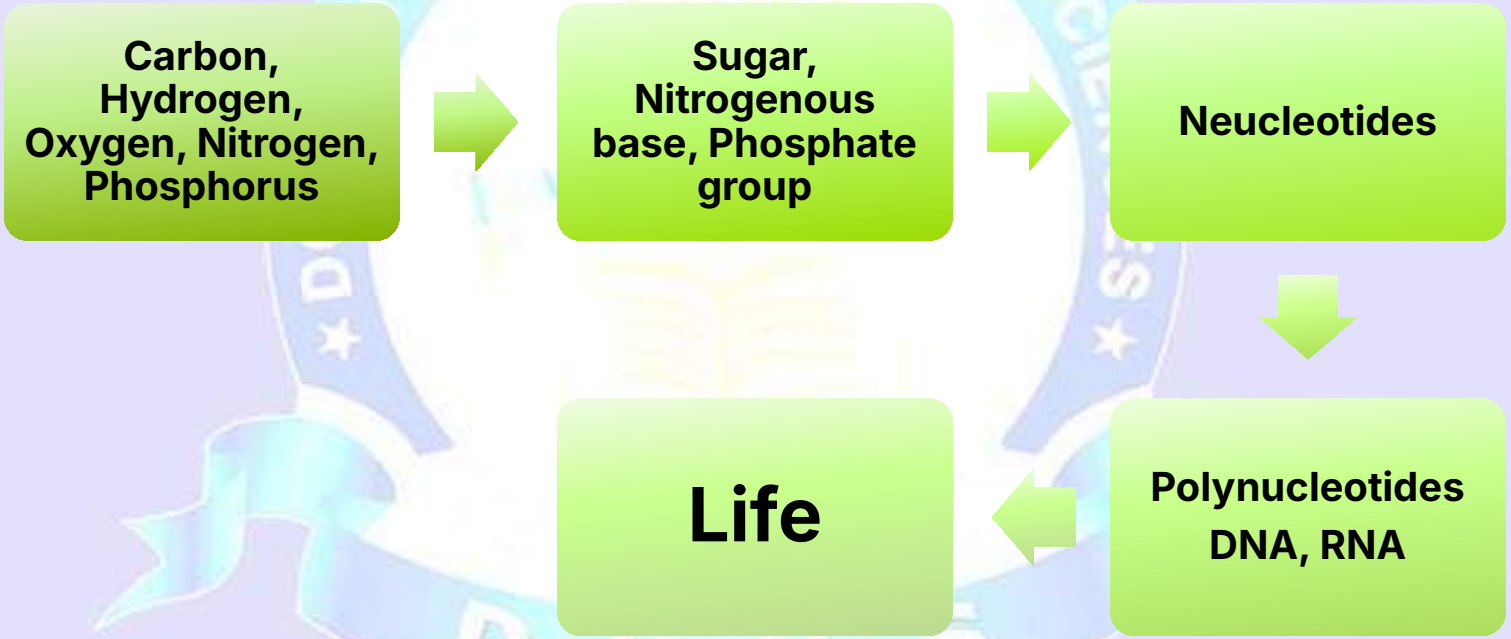
- The molecules interact to form tissues
- Tissues perform various functions



CHEMICAL ORGANIZATION



Examples Chemical Organization



Examples Chemical Organization



Examples Chemical Organization



Macromolecules

- Carbon
- Hydrogen
- Oxygen
- Nitrogen
- Phosphorus

- Carbohydrates
- Proteins
- Lipids
- Nucleic Acids

CHEMICAL ORGANIZATION IN THE HUMAN BODY

- Water and Minerals
- Chemicals within the body are classified as inorganic or organic
  - Inorganic compounds usually do not have carbon as the major structural element, and ionic bonds are common.
- Organic Compounds
  - Organic compounds contain carbon, hydrogen, oxygen, nitrogen, Phosphorus and sometimes other trace elements like sulphur, iron, zinc etc. Covalent bonds are most common

CHEMICAL ORGANIZATION IN THE HUMAN BODY

- Carbohydrates
  - Carbohydrates are sugars and starches. They are important energy sources for metabolic operations.
  - A simple sugar, or monosaccharide, has from three or more carbon atoms. Glucose is the most important monosaccharide.
  - Disaccharides and polysaccharides formed by dehydration synthesis. They can be broken apart through hydrolysis
  - Glycogen is the most important polysaccharide in our bodies; it is a storage form of glucose.
  - Hetero-polysaccharides have any other molecule along with CHO

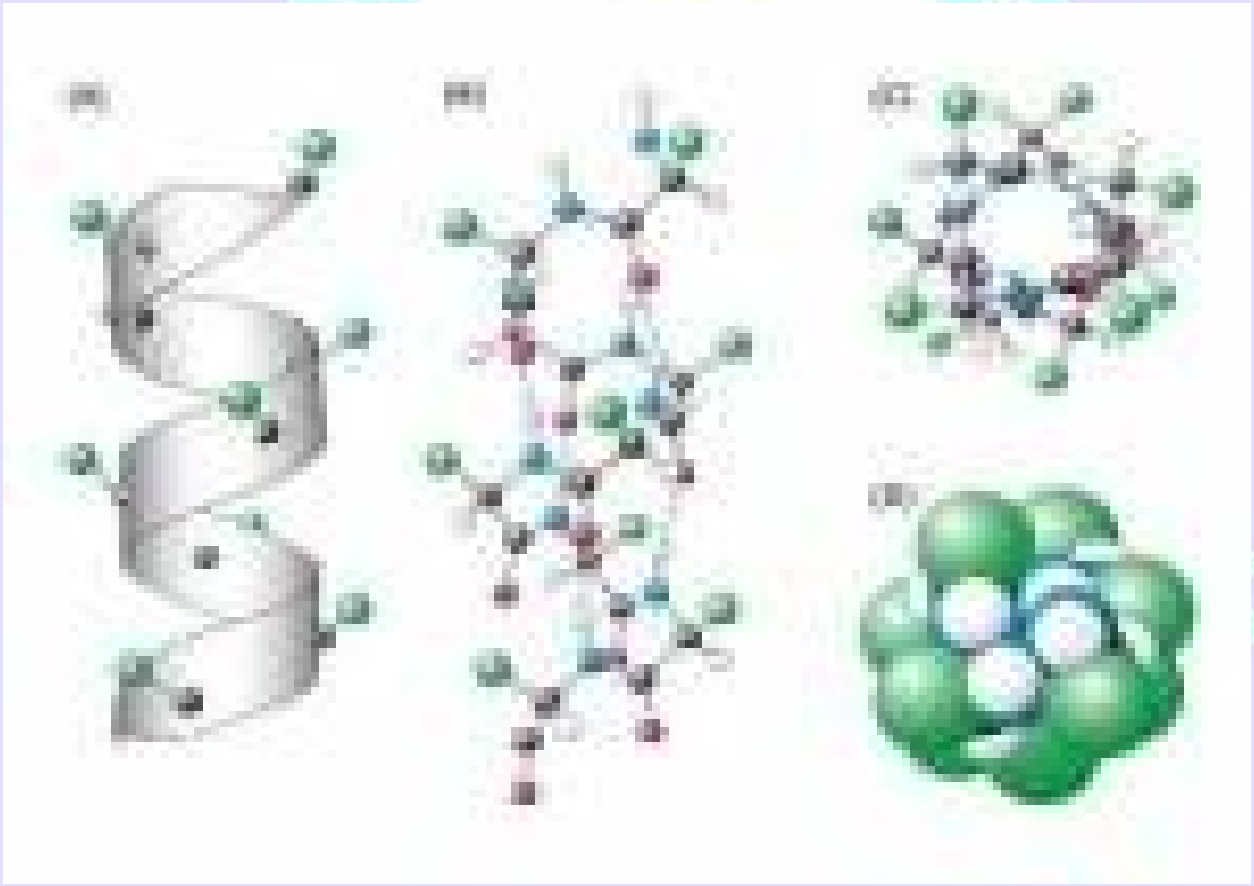
CHEMICAL ORGANIZATION IN THE HUMAN BODY (Cont..)

- Lipids
  - Lipids are fats, oils, and waxes
  - Fatty acids are important energy sources
  - Fatty acids can be linked to a glycerol to form triglycerides, an important storage form for lipids
  - Prostaglandins and some steroids function as hormones
  - Cholesterol is a precursor of steroid hormones, and is an important component of membranes Cholesterol, phospholipids, and glycolipids are the structural lipids of the body

CHEMICAL ORGANIZATION IN THE HUMAN BODY

- **Proteins**
  - Proteins are composed of amino acids linked by peptide bonds.
  - Structural proteins contain twenty different amino acids
  - The structure of a protein depends upon the amino acid sequence, and interactions between the amino acids, the surrounding fluid; and other protein molecules.
  - Proteins are hormones, regulators of metabolic activity, and buffers.
  - They also function in movement, transportation of dissolved materials, and defense against disease and physical stresses.
  - Enzymes are proteins that lower the activation energy requirements for chemical reactions.
  - Other special proteins include glycoproteins and proteoglycans.
  - Proteins and large molecules in solution create a colloid.
  - Amino acids and proteins in body fluids normally carry a negative electrical charge.

PROTEIN STRUCTURE



CHEMICAL ORGANIZATION IN THE HUMAN BODY

- **Nucleic Acids**
  - Nucleic acids store and process genetic information concerning the structure of proteins. There are two types of nucleic acids, ribonucleic acid (RNA) and deoxyribonucleic acid (DNA)
  - Nucleic acids are chains of nucleotides. Each nucleotide contains a phosphate group, a sugar, and a nitrogen base
  - In RNA, the sugar is ribose, and the nitrogen base may be adenine, guanine, cytosine, or uracil
- **Other Organic Compounds**
  - High-energy compounds store the energy released during the catabolism of other compounds
  - ATP, adenosine triphosphate, is the most important high-energy compound
  - Vitamins are essential nutrients; many of them act as coenzymes for enzymes
  - There are fat-soluble and water-soluble vitamins with distinct requirements for absorption and methods of storage and excretion
  - Porphyrins are important in transporting oxygen and assisting with energy transfer and capture

THE LIFE

Function	Organ	Chemical
Food Utilization (Metabolism) Digestion/Absorption	GIT, Liver....	Hormones, Enzymes, ATP
Respiration	Lungs	O_2, CO_2
Excretion	Kidneys, Liver, Lungs	Hormones, Enzymes, Waste chemicals
Transport	Blood/plasma	Water
Cell to cell communication	Receptors	Proteins
Thinking Process	Brain	Neurochemical (transmitters/ peptides)
Reproduction/ cell division	Sex organs/Cells	DNA→RNA→Proteins
Smile, Care, Love	Heart	The neurochemicals and local hormones

Thank you